

Comparative analysis of radioholographic methods of processing radio occultation data

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Abstract. Two methods for the determination of refraction angle profiles from radio occultation measurements in multipath areas are analyzed and compared: (1) the radio-optic method based on the analysis of the local spatial spectra of the measured wave field and (2) the back propagation of the received wave field to a single-ray region. The basic limitations of the radio-optic method are (1) the restriction of the resolution of refraction angle profiles due to the uncertainty relation of refraction angle and impact parameter and (2) diffractive effects in subcaustic areas, where the spatial spectra cannot be interpreted in terms of geometric optical rays. The basic limitation of the back propagation method is related to ray and caustic structures, which may not contain single-ray areas. It is shown that strong refraction reduces the uncertainties of refraction angle and impact parameter. On the other hand, strong refraction or superrefraction is responsible for complicated caustic structures that cannot be resolved by the back propagation technique. The two methods are thus complementary to each other and can be combined for processing lower tropospheric occultation data. This analysis is corroborated by numerical simulations based on global fields of atmospheric variables from analyses of the European Centre for Medium Range Weather Forecast.

1. Introduction

Radio occultation sounding of the Earth's atmosphere using the radio signals of the Global Positioning System (GPS) can be a valuable source of information about key atmospheric variables [Ware *et al.*, 1996; Rocken *et al.*, 1997; Kursinski *et al.*, 1997]. The experience of processing data from the Microlab-1 satellite as well as theoretical considerations show that the most complicated problems of the interpretation of radio occultation data arise in the lower troposphere [Gorbunov *et al.*, 1996b; Gorbunov and Gurvich, 1998a, 1998b]. On the other hand, lower tropospheric data are of the greatest importance for numerical weather prediction (NWP). The difficulties of data processing can be classified as follows:

1. They can be classified as algorithmic problems. Processing the phase and amplitude measurements in multipath areas requires a special technique for the separation of interfering rays.

2. They can be classified as technical problems. The receiver used in the Microlab-1 satellite was not capable of tracking the signal with the strong amplitude and phase scintillations resulting from lower tropospheric inhomogeneities and multipath propagation, on the background of significant refractive attenuation of the signal. This resulted in the lack of reliable lower tropospheric data.

Two different approaches to processing radio occultation data in multipath areas have been discussed in the literature: (1) the analysis of spatial spectra, which we will refer to as the radio-optic method [Lindal *et al.*, 1987; Pavelyev, 1998], and (2) the back propagation technique, which is also referred to as diffraction correction [Marouf *et al.*, 1986; Gorbunov *et al.*, 1996b; Karayel and Hinson, 1997; Gorbunov

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and Gurvich, 1998a, 1998b]. Both methods can be termed radioholographic, because they are based on the analysis of records of complex radio signals, or radio holograms. In this paper, we analyze their capabilities in processing data of GPS radio occultation sounding of the Earth's atmosphere.

Because of the lack of reliable lower tropospheric data our analysis can only be based on simulated radio occultations. The simulations were performed with global fields of pressure, temperature, and humidity from analyses of the European Centre for Medium-Range Weather Forecasts (ECMWF).

2. Radio Occultation Technique for Sounding Planetary Atmospheres

The radio occultation technique is based on the transillumination of a planetary atmosphere with radio waves emitted by a spaceborne transmitter and received by a spaceborne receiver. In the sounding of the Earth's atmosphere the signal is transmitted by a GPS satellite and received by a low-Earth orbiter (LEO) equipped with a GPS receiver. The typical observation geometry is shown in Figure 1. During a (setting) occultation the satellites are moving in such a way that the radio ray connecting them immerses into the atmosphere until the receiver reaches the solid Earth's shadow or loses the signal earlier. The phase (or Doppler frequency) and amplitude of the received signal are recorded for their subsequent use for the reconstruction of atmospheric refractive index, which can be interpreted in terms of atmospheric variables [Gorbunov et al., 1996a; Ware et al., 1996; Rocken et al., 1997; Kursinski et al., 1997; Steiner et al., 1999].

Basic approximations for the interpretation of radio occultation data are geometric optics and local

spherical symmetry, which allows for the formulation of the inverse problem in terms of measured dependence of refraction (bending) angle ε versus impact parameter p . The vertical profile of atmospheric refractive index $n(r)$ must be reconstructed. In a spherically symmetrical medium, rays are plane curves, and impact parameter is a ray invariant such that at each point of the ray, $p = n(r)r \sin \psi = n(r)|\mathbf{r} \times \mathbf{u}|$, where ψ is the angle between local ray direction $\mathbf{u}$ and radius vector $\mathbf{r}$. This statement is referred to as Snell's law for a spherically symmetric medium or the formula of Bouger [Born and Wolf, 1964]. Using the relation between the measured Doppler frequency ω_{LEO} and the frequency of the transmitted signal ω_{GPS} , complemented with the vector form of Snell's law, we can write equations for the determination of ray directions $\mathbf{u}_{\text{GPS}}$ and $\mathbf{u}_{\text{LEO}}$ [Fjeldbo et al., 1971; Vorob'ev and Krasil'nikova, 1994]:

$$\frac{c - \mathbf{V}_{\text{LEO}} \cdot \mathbf{u}_{\text{LEO}}}{c - \mathbf{V}_{\text{GPS}} \cdot \mathbf{u}_{\text{GPS}}} \sqrt{\frac{c^2 - V_{\text{GPS}}^2}{c^2 - V_{\text{LEO}}^2}} = \frac{\omega_{\text{LEO}}}{\omega_{\text{GPS}}}, \quad (1)$$

$$\mathbf{r}_{\text{LEO}} \times \mathbf{u}_{\text{LEO}} = \mathbf{r}_{\text{GPS}} \times \mathbf{u}_{\text{GPS}},$$

where c is the light velocity in a vacuum, $\mathbf{V}_{\text{GPS}}$ and $\mathbf{V}_{\text{LEO}}$ are the GPS and LEO velocities and $\mathbf{r}_{\text{GPS}}$ and $\mathbf{r}_{\text{LEO}}$ are the GPS and LEO positions. It is assumed that both satellites are in a vacuum.

The refraction angle can be calculated as angle ε between $\mathbf{u}_{\text{GPS}}$ and $\mathbf{u}_{\text{LEO}}$. Impact parameter p equals $|\mathbf{r}_{\text{LEO}} \times \mathbf{u}_{\text{LEO}}| = |\mathbf{r}_{\text{GPS}} \times \mathbf{u}_{\text{GPS}}|$. Dependence $\varepsilon(p)$ can be inverted using the Abel integral transform [Phinney and Anderson, 1968; Fjeldbo et al., 1971], which is the exact solution for a spherically symmetrical field $n(r)$ but is also used as an approximation

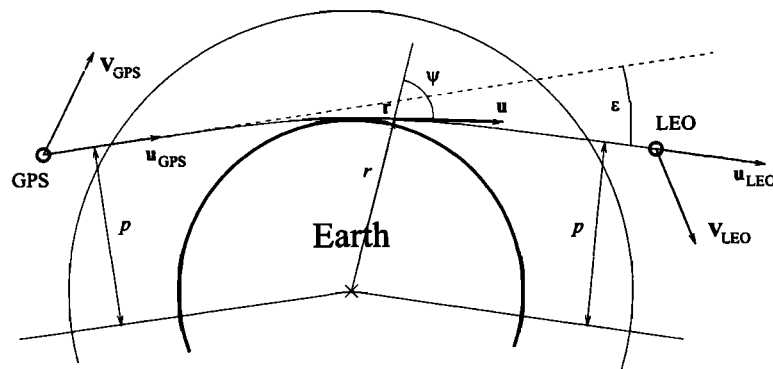


Figure 1. Radio occultation geometry.

for real atmospheres with horizontal gradients of n .

Because Snell's law (the formula of Bouger) is violated in the real atmosphere with horizontal gradients ($|\mathbf{r}_{\text{LEO}} \times \mathbf{u}_{\text{LEO}}| \neq |\mathbf{r}_{\text{GPS}} \times \mathbf{u}_{\text{GPS}}|$), ε and p determined from (1) will not coincide with the real bending angle and impact parameter (the latter will be different at GPS and LEO) [Ahmad and Tyler, 1999]. Nevertheless, the dependence $\varepsilon(p)$ can be looked at as a known functional of the refractive index field and as such assimilated into global atmospheric circulation models using three-dimensional or four-dimensional variational assimilation techniques (3DVar or 4DVar) [Eyre, 1994; Gorbunov et al., 1996a]. In this approach, (1) is looked at as the definition of the functional that can be applied without making the assumption of the spherical symmetry of the refracting medium. This approach avoids errors due to the approximation of the local spherical symmetry.

Equation (1) assumes single-ray propagation. A typical situation for the lower troposphere is, however, multipath propagation. Doppler frequency can be computed as the derivative of the phase of the wave field, also in a multipath area. Formal application of (1) results then in a characteristic oscillating pattern in the (ε, p) plane, which reflects the interference oscillations of the phase [Gorbunov and Gurvich, 1998b]. The resulting dependence $\varepsilon(p)$ proves ambiguous and cannot be used in inversion algorithms (inversion is usually preceded by some heuristic procedure of the elimination of ambiguities). We come thus to the problem of the determination of the ray structure of a wave field measured by a spaceborne GPS receiver.

3. Radio-optic Method

The radio-optic method was used for processing soundings of planetary atmospheres [Lindal et al., 1987], and recently its application to GPS radio occultation data was suggested [Pavelyev, 1998]. The method is based on the analysis of the local spatial spectra of the measured complex wave field. Local spectra are computed by Fourier analysis of the record of the complex field in small sliding apertures. Rays must correspond to maxima in the Fourier spectra. The limitations of this method will now be analyzed.

We shall use Cartesian coordinates (x, z) with the x axis pointing in the direction of the initial wave propagation and the z axis transverse to it. The

$(0,0)$ point of the coordinate frame is located in the Earth's local curvature center.

The wave field is observed at distance x from the Earth's limb, which is estimated as $\sqrt{r_{\text{LEO}}^2 - r_E^2}$, where r_{LEO} is the LEO orbit radius and r_E is the Earth's local curvature radius. The spatial spectrum of the wave field is analyzed inside vertical apertures with size A . The angular resolution of the spatial spectrum can be estimated as λ/A . Assuming that the wave field can be described in the geometric optical approximation, we shall interpret this relationship in terms of refraction angles and impact parameters.

Each ray can be associated with a plane wave $\exp(i\sqrt{k^2 - k_z^2}x + ik_z z)$. Using the approximation of small refraction angle and infinitely remote GPS, we can write the following relationships for the refraction angle and impact parameter at observation point (x, z) :

$$\varepsilon = \frac{k_z}{k}, \quad (2)$$

$$p(x, z, \varepsilon) = z + x\varepsilon.$$

The second relationship can be rewritten as follows:

$$\varepsilon_G(x, z, p) = \frac{p - z}{x}, \quad (3)$$

which specifies the geometric relationship for the refraction angle and impact parameter of a ray coming to the observation point. In order to find all such rays we must solve the following equation:

$$\varepsilon_G(x, z, p) = \varepsilon(p). \quad (4)$$

The spectral resolution, $\Delta k_z = 2\pi/A$, determines the resolution of the refraction angle:

$$\Delta\varepsilon = \frac{\Delta k_z}{k} = \frac{\lambda}{A}. \quad (5)$$

On the other hand, the aperture size determines the resolution of the impact parameter:

$$\Delta p = \left| \frac{dp}{dz} \right| A = \frac{A}{\zeta}, \quad (6)$$

$$\zeta = \left| 1 - x \frac{d\varepsilon}{dp} \right|.$$

Because refraction angle and impact parameter must satisfy geometric relationship (3), their uncertainties can be estimated as follows:

$$\begin{aligned}\delta p^2 &= \Delta p^2 + x^2 \Delta \varepsilon^2 = \frac{A^2}{\zeta^2} + \frac{x^2 \lambda^2}{A^2}, \\ \delta \varepsilon^2 &= \Delta \varepsilon^2 + \frac{\Delta p^2}{x^2} = \frac{\lambda^2}{A^2} + \frac{A^2}{x^2 \zeta^2}.\end{aligned}\quad (7)$$

These equations allow for the estimation of the optimal aperture size that minimizes both $\delta \varepsilon$ and δp :

$$A = \sqrt{\zeta x \lambda}. \quad (8)$$

The uncertainties of the refraction angle and impact parameter can then be estimated by their order of magnitude:

$$\delta \varepsilon \approx \sqrt{\frac{\lambda}{\zeta x}}, \quad (9)$$

$$\delta p \approx \sqrt{\frac{x \lambda}{\zeta}}. \quad (10)$$

We have thus the uncertainty relation

$$\delta \varepsilon \delta p \sim \frac{\lambda}{\zeta}. \quad (11)$$

Because $d\varepsilon dp$ is the natural measure in the space of (straight-line) rays [Santalo, 1976], we can understand this relationship as the estimation of the volume of a quantum cell in this space. Two rays can be distinguished if they cannot be covered with one cell.

For the GPS radio occultations performed by the Microlab 1 satellite the observation distance was ~ 3000 km, and the wavelength ~ 20 cm. For weak refraction ($\zeta \approx 1$) the uncertainties of refraction angle and impact parameter are then estimated as 2.6×10^{-4} rad and 800 m, respectively. For the purposes of NWP we must reconstruct pressure with an accuracy of 1 mbar (0.1%), temperature with an accuracy of 1 K (0.3%), and geopotential heights with an accuracy of 10 m. The accuracy of refraction angle in the lower troposphere is then estimated as 0.1%, or $\sim 3 \times 10^{-5}$ rad. The required accuracy of impact parameter must be ~ 10 m. Although the above estimations of the uncertainties exceed these values, we must not forget that multipath propagation is only

significant in the lower troposphere, where ζ can be fairly large (30–300), and the uncertainties will be reduced by a factor of $\sqrt{\zeta}$. It must also be noted that the amplitude of the signal in the geometric optical approximation is also inversely proportional to $\sqrt{\zeta}$.

In the lower troposphere, where $\zeta \gg 1$, it can be approximately written $\zeta \approx |x d\varepsilon/dp|$. Parameter ζ is then proportional to the observation distance x , and from (9) it follows that the uncertainty of the refraction angle can be reduced by means of the forward propagation of the observed wave field to a plane located at a larger distance from the atmosphere. As seen from (10), however, this will not improve the uncertainty of the impact parameter, and thus the limitation of the vertical resolution will still persist. This shows that the overall uncertainty volume $\delta \varepsilon \delta p$ is not so important as its shape.

The uncertainty relation is not the only restriction of the radio-optic method. Because it is based on the representation of the wave field in the form of a sum of waves corresponding to geometric optical rays, this method is incapable of taking diffraction effects into account.

The relation between the geometric optical approximation and the exact solution of the diffractive problem was already discussed in literature [Kravtsov and Orlov, 1990], and it was shown that these solutions are very close to each other everywhere in a free space except subcaustic zones and the size of a subcaustic zone was estimated. The approach based on the theory of canonic operators [Mishchenko et al., 1978] results in the same conclusion. This is also corroborated by the numerical simulations described in Section 5. Because the wave field in subcaustic zones cannot be interpreted in terms of geometric optical rays, the radio-optic method cannot be applied there.

As we shall see in Section 5, subcaustic zones (for GPS/MET observation geometry) are narrow enough to be excluded without resulting in a significant loss of data. On the other hand, the caustic structure is a priori unknown (knowledge of the caustic structure is equivalent to knowledge of $\varepsilon(p)$). At the moment we cannot formulate the algorithm of the identification of subcaustic zones for their exclusion.

4. Back Propagation Method

The back propagation method was first suggested for the profiling of Saturn's rings from diffracted images [Marouf et al., 1986]. The method was initially designed to correct for the diffraction effects,

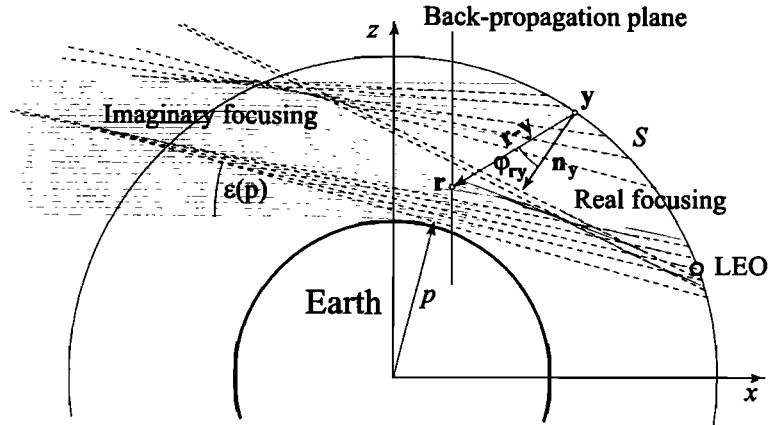


Figure 2. Multipath propagation geometry and back propagation. Nonmonotonous vertical profile $\varepsilon(p)$ results in real focusing of rays and multipath propagations. Back propagation of the wave field (back-propagated rays are shown as dashed lines) results in imaginary focusings. Back-propagated field at point $\mathbf{r}$ is calculated via diffractive integral (12) as a superposition of infinitesimal fields from the points of curve S . Each point $\mathbf{y}$ can be represented as a transmitter with intensity $u_0(\mathbf{y})$ and direction diagram $\cos \varphi_{\mathbf{r}\mathbf{y}}$.

which were significant in the interpretation of images of Saturn's rings recorded from a large observation distance.

Recently the method was modified for processing radio occultation data in multipath areas [Gorbunov et al., 1996b; Karayel and Hinson, 1997; Hinson et al., 1997; Hison et al., 1998; Gorbunov and Gurvich, 1998a; Gorbunov and Gurvich, 1998b]. The method is based on the solution of the boundary problem for the Helmholtz equation in a vacuum, the boundary conditions being delivered by the measurements of the complex field along the LEO orbit. From the geometric optical viewpoint, this solution corresponds to the back continuation of the rays as straight lines. Such a continuation conserves both ray direction (and therefore the refraction angle) and impact parameter. The field is back-propagated to some plane positioned, if possible, in a single-ray area (Figure 2).

The new feature of the method is that although the back propagation is performed as in a vacuum, the back propagation plane can (and must) be positioned inside the atmosphere, and the back-propagated field is thus not the real wave field at the location of the back propagation plane.

Assuming that the source of the wave field (GPS) is stationary and that the LEO orbit is located in a vertical plane, which we will refer to as the occultation plane, the back-propagated field is calculated

using the diffractive integral [Vladimirov, 1971; Gorbunov et al., 1996b]:

$$u(\mathbf{r}) = \left(\frac{k}{2\pi} \right)^{1/2} \int_S u_0(\mathbf{y}) \times \cos \varphi_{\mathbf{r}\mathbf{y}} \frac{\exp(-ik|\mathbf{r} - \mathbf{y}| + i\pi/4)}{|\mathbf{r} - \mathbf{y}|^{1/2}} dS_{\mathbf{y}}, \quad (12)$$

where $\varphi_{\mathbf{r}\mathbf{y}}$ is the angle between vector $\mathbf{r} - \mathbf{y}$ and normal $\mathbf{n}_{\mathbf{y}}$ to curve S at current integration point $\mathbf{y}$, and $u_0(\mathbf{y})$ is the boundary condition. Curve S is the LEO trajectory. Although real occultations are not vertical, it is possible to reduce the problem to the case where the GPS satellite is stationary, and the LEO satellite is moving vertically, by means of an appropriate coordinate transform [Gorbunov et al., 1996b].

As discussed in Section 3, diffractive effects in single-ray areas are negligible. Therefore, if the back propagation plane is positioned in a single-ray area, then the back-propagated field can be interpreted in terms of geometrical optics. The Doppler frequency is calculated by means of the numerical differentiation of its phase. The refraction angles and impact parameters are then found by solving (1).

The calculation of Doppler frequency by means of the numerical differentiation of the phase can be performed with a much smaller effective aperture than that required for the radio-optic method. The

wavenumber-aperture uncertainty relation does not apply here because of the use of a very strong assumption of single-ray propagation. This can be explained as follows: Given a complex wave field $u(z)$, the uncertainty relation states that $\delta k_z \delta z \geq 2\pi$, where δz is the aperture size and δk_z is the spectral resolution. If the spatial spectrum is grouped around some frequency $k_z^{(0)}$, then for an aperture size approximately equal to the oscillation period of the wave field, $2\pi/k_z^{(0)}$, the spatial spectrum cannot be determined, because $\delta k_z \geq k_z^{(0)}$. The situation is different if it is a priori known that the field is measured in a single-ray area. It can then be represented in the form $u(z) = a(z) \exp [i \int k_z(z) dz]$, where amplitude $a(z)$ and spatial frequency $k_z(z)$ are very slowly changing inside an oscillation period. In this case the frequency can be accurately calculated from the oscillation period, and wave vector $\mathbf{k}(z) = [\sqrt{k^2 - k_z^2(z)}, k_z(z)]$ points in the direction of the single ray at a given point.

The basic restriction of the back propagation method is related to the configuration of rays and caustics of the back-propagated field in the occultation plane, which will now be investigated. The straight line associated with a ray characterized by refraction angle ε and impact parameter p is described by the following equation:

$$x \sin \varepsilon(p) + z \cos \varepsilon(p) = p. \quad (13)$$

A caustic is the envelope of the family of rays. In the simplest cases each caustic is the boundary between multipath and single-ray propagation areas. The equation for caustics is derived by complementing (13) with its derivative with respect to parameter p [Korn and Korn, 1968]. Solving the resulting system, we arrive at the equations for caustics:

$$\begin{aligned} x &= p \sin \varepsilon(p) + \frac{\cos \varepsilon(p)}{\varepsilon'_p(p)}, \\ z &= p \cos \varepsilon(p) - \frac{\sin \varepsilon(p)}{\varepsilon'_p(p)}. \end{aligned} \quad (14)$$

Assuming that $|\varepsilon'_p(p)| < \infty$, we see that the complete envelope can be represented as a set of continuous caustics corresponding to the monotonous fragments of the dependence $\varepsilon(p)$. They can be subdivided into two classes: real caustics with $\varepsilon'_p(p) > 0$

arising because of focusing of rays in the observation plane, and imaginary caustics with $\varepsilon'_p(p) < 0$ arising because of defocusing (or imaginary focusing).

Equation (14) indicates that each ray with a p such that $\varepsilon'_p(p) \neq 0$ touches only one caustic (real or imaginary according to the sign of $\varepsilon'_p(p)$). The perigee points of the rays defined by the equations

$$\begin{aligned} x &= p \sin \varepsilon(p), \\ z &= p \cos \varepsilon(p) \end{aligned} \quad (15)$$

are located between real and imaginary caustics, and thus they are in a single-ray area. This suggests back-propagating the measured wave field to this curve. This solution is, however, associated with algorithmic difficulties, because the dependence $\varepsilon(p)$ itself must yet be determined, and iterations are thus necessary. Additional difficulties will also result from the complicated shape of such a curve. Currently, we back-propagate the measured field to a vertical straight line, whose position x can be estimated from (15) as $x = r_E \varepsilon_{\max}$ where the maximum refraction angle can be estimated from the processing of the measurements without back propagation. In most cases this estimation is sufficient for disentangling multipath ray structures. Other practical criteria for the right choice of the back propagation plane position can be the following.

1. There are minimum amplitude fluctuations of back-propagated field [Karayel and Hinson, 1997]. This criterion is based on the fact that the strongest amplitude fluctuation arise due to interference in multipath areas.

2. There is a minimum difference of optical path in L1 and L2 frequency channels. This criterion uses the strong frequency dependence of interference effects in multipath areas, while in one ray areas the frequency dependence of optical path can only be due to diffraction effects, which are very weak in single-ray areas for GPS/MET observations.

The situation becomes more complicated in the presence of a superrefraction layer. In this case the geometric optical dependence $\varepsilon(p)$ is characterized by a very sharp peak, where formally $|\varepsilon'_p(p)| = \infty$. As seen from (14), the caustic structure will then contain real and imaginary caustics very close to each other. In such cases the back propagation plane will always have intersections with caustics. Refraction angle profiles with sharp spikes caused by superrefraction

or conditions approaching superrefraction constitute the basic limitation of the back propagation method.

As we have already seen, however, strong refraction resulting in large values of ζ is most favorable for the radio-optic method. This suggests combining the two methods for processing lower tropospheric data.

5. Numerical Simulations

In the numerical simulations we compared the results of processing artificial occultation data using the back propagation and radio-optic methods. The artificial data were generated using the split-step numerical solution of the diffraction problem in the atmosphere [Martin, 1992; Karayel and Hinson, 1997; Gorbunov and Gurvich, 1998b]. The model of the atmospheric refractive index field was based on the global fields of temperature, humidity, and pressure from analyses of ECMWF. The occultation geometry was taken from real GPS/MET occultations and transformed to vertical occultation planes. The analysis fields are given with a time step of 6 hours; the analysis nearest to the time of the GPS/MET occultation event was chosen.

Using the geometric optical solution, we calculated the caustic structure, which allowed for the determination of the optimal position of the back propagation plane. The radio-optic method was applied to the complex field forward-propagated to a plane located at $x = 5000\text{km}$, which allowed for a slight improvement of the resolution. The forward-propagation plane was covered by sliding apertures with a size of $\sqrt{\zeta x \lambda}$. Factor ζ was evaluated from (6), where $\varepsilon(p)$ was computed from the observed phase with a very strong smoothing. In our simulation the accuracy of this estimation was sufficient. The basic sliding step of the sliding apertures was $\sim 1.5\text{ km}$.

Inside an aperture centered at z_0 the (forward propagated) complex field $u(z)$ was represented in the form

$$\begin{aligned} u(z) &= a(z) \exp(i\varphi(z)) \\ \varphi(z) &= \alpha_0 + \alpha_1(z - z_0) + \alpha_2(z - z_0)^2 + \Delta\varphi(z) \end{aligned} \quad (16)$$

where $a(z)$ is the amplitude, $\varphi(z)$ is the accumulated phase, and α_i are the regression coefficients. The complex field $a(z) \exp[i\Delta\varphi(z)]$ with downconverted

spatial frequency was subjected to the Fourier analysis. This procedure approximately corresponds to the representation of the wave field $u(z)$ as a sum of spherical waves, and the square phase term in (16) approximately corrects for the wave front curvature. The discretization step Δz was chosen to be equal to $\lambda/\Delta\varepsilon \approx 200\text{ m}$, where $\Delta\varepsilon = 0.001\text{ rad}$ is the a priori estimation of the width of the angular spectrum.

Downconverted frequencies σ_z correspond to spatial frequencies $k_z = \alpha_1 + \sigma_z$ of the initial field $u(z)$ in the center of the aperture. The wave vector is then equal to $\mathbf{k} = (k_x, k_z) = (\sqrt{k^2 - k_z^2}, k_z)$. Using Snell's law, we can compute refraction angles $\varepsilon(x, z_0, k_z)$ and impact parameters $p(x, z_0, k_z)$ corresponding to the ray direction of wave vector $\mathbf{k}$. This defines a mapping of (z_0, k_z) to (ε, p) . For fixed observation point (x, z_0) , dependencies $\varepsilon(x, z_0, k_z)$ and $p(x, z_0, k_z)$ are the parametric representation of curve $\varepsilon_G(x, z_0, p)$ in the (ε, p) plane, where k_z is the curve parameter. Each local spatial spectrum $\tilde{u}(x, z_0, k_z)$ can then be plotted along this curve in (ε, p) coordinates. For all possible z_0 , curves $\varepsilon_G(x, z_0, p)$ cover the complete (ε, p) plane, and the maxima of the spectra must thus trace the dependence $\varepsilon(p)$.

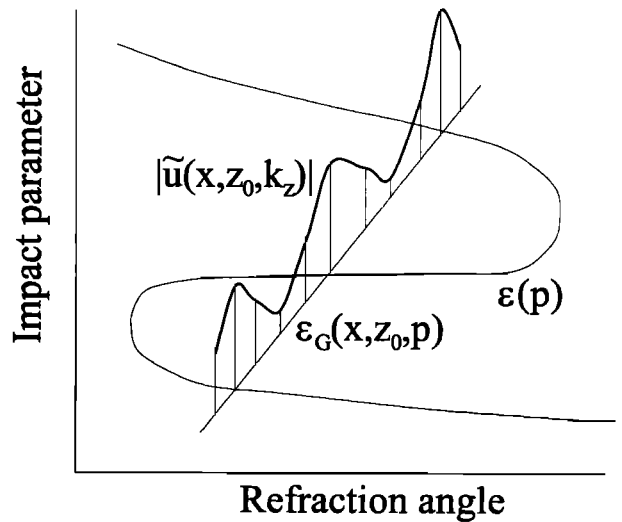


Figure 3. Geometry of the calculation of spatial spectra. Each ray is represented as a point in the (ε, p) plane. Each observation point (x, z_0) is represented as curve $\varepsilon_G(x, z_0, p)$. Each spatial frequency k_z corresponds to point $(\varepsilon(x, z_0, k_z), p(x, z_0, k_z))$ of this curve. Rays coming to given point are defined by the equation $\varepsilon_G(x, z_0, p) = \varepsilon(p)$. Absolute values of the spatial spectra of wave field $|\tilde{u}(x, z_0, k_z)|$ must have maxima corresponding to rays.

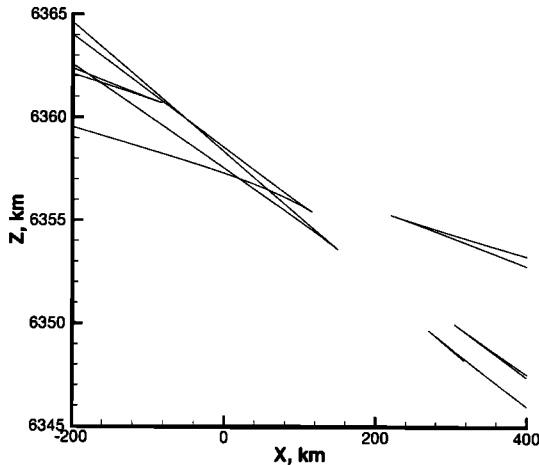


Figure 4. Simulated occultation event 0478 (February 2, 1997, 1637 UTC) at 29.3°S 173.0°E: caustic structure of back-propagated wave field.

This is schematically shown in Figure 3, where dependencies $\varepsilon_G(x, z_0, p)$ and $\varepsilon(p)$ have three intersections, i.e. three rays interfere at a given observation point, and three maxima must be expected for absolute values of spatial spectrum $|\tilde{u}(x, z_0, k_z)|$.

An example of processing artificial occultation data using the back propagation and radio-optic methods is shown in Figures 4 and 5. We computed the geometric optical refraction angles $\varepsilon_{GO}(p)$ by means of

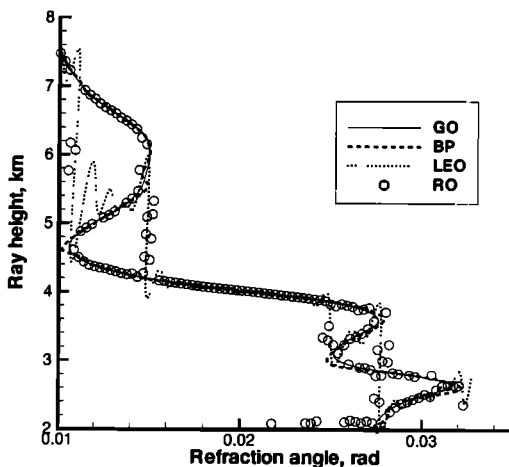


Figure 5. Simulated occultation event 0478 (February 2, 1997, 1637 UTC) at 29.3°S 173.0°E: refraction angle profiles (1) geometric optical (GO), (2) from back-propagated field (BP), (3) from phase measured at LEO, and (4) spectral maxima from radio-optical analysis (RO).

the standard ray-tracing technique [Gorbunov et al., 1996a]. Using them as the exact solution, we validate the back propagation and the radio-optic methods. Figure 4 shows the configuration of caustics of the back-propagated wave field. The back propagation plane can be located between the real and imaginary caustics at $x = 200$ km. For the radio-optic analysis, factor ζ was estimated at 60, which corresponds to aperture size $A = 8$ km. Figure 5 shows refraction angle profiles: (1) geometric optical $\varepsilon_{GO}(p)$, (2) $\varepsilon_{BP}(p)$ computed from back-propagated field, (3) $\varepsilon_{LEO}(p)$ computed from phase measured at LEO, and (4) spectral maxima from radio optical analysis. Instead of impact parameter we use the ray leveling height defined as $p - r_E$.

The multipath regions are marked with strong oscillations of the dependence $\varepsilon_{LEO}(p)$, which becomes ambiguous there. The oscillations are concentrated near the branch of the dependence $\varepsilon_{GO}(p)$ corresponding to the strongest of the interfering rays. Spectral maxima computed in the radio-optic method show a good agreement with $\varepsilon_{GO}(p)$ everywhere, except in relatively narrow vicinities of the caustics, where diffraction effects are significant and spectral maxima cannot be interpreted in terms of rays. The best agreement with the exact solution is indicated by the back propagation method.

Figures 6 and 7 show another example of an artificial radio occultation, where the back propagation method is not so successful in the lowest troposphere. The caustic structure shown in Figure 6 is very characteristic for a superrefraction layer. No positioning

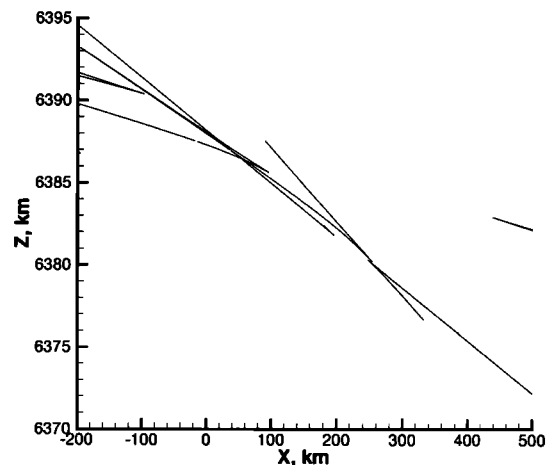


Figure 6. Simulated occultation event 0681 (February 2, 1997, 2342 UTC) at 29.1°S 171.4°W: caustic structure of back-propagated wave field.

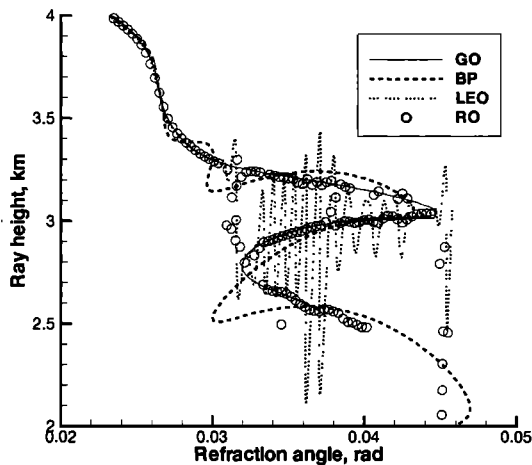


Figure 7. Simulated occultation event 0681 (February 2, 1997, 2342 UTC) at 29.1°S 171.4°W: refraction angle profiles (1) geometric optical (GO), (2) from back-propagated field (BP), (3) from phase measured at LEO, and (4) spectral maxima from radio-optical analysis (RO).

of the back propagation plane can get rid of multipath propagation. For the radio-optic method, the value of ζ was estimated to be 215, which corresponds to aperture size $A = 15$ km.

Figure 7 shows refraction angles. The vertical profile of refraction angles is characterized by a sharp spike at height of ~ 3 km. This spike is responsible for the real and imaginary caustic with overlapping projections to the x axis seen in Figure 6. For ray heights below 3.5 km the back propagation method proves incapable of reconstructing the true ray structure. The dependence $\varepsilon(p)$ is ambiguous here. This ambiguity, however, can serve as the indication that the back propagation plane intersects caustics and the back propagation method is inapplicable. Such a situation can also be diagnosed by the amplitude fluctuations [Karayel and Hinson, 1997]. For this occultation the radio-optic method yields a much better agreement of the reconstructed ray structure with the exact geometric optical solution. However, the problem of extra spectral maxima in subcaustic areas remains unsolved.

6. Conclusions

The presented analysis allows for the following comparative characterization of the back propagation and radio-optic methods. The radio-optic method has two basic limitations.

1. One limitation is the uncertainty relation of refraction angle and impact parameter. The uncertainties, however, decrease in the lowest troposphere under the conditions of strong refraction, where the limitation of the resolution becomes less significant and thus the biggest impact of this method is expected.

2. Diffractive effects in subcaustic zones result in “false” maxima of the spatial spectra of the wave field, i.e., maxima that cannot be interpreted in terms of rays. Sorting out these maxima can become an important problem, especially with complicated profiles of refraction angles which can be expected in the lower troposphere.

The basic limitation of the back propagation method is related to complicated caustic structures, which can arise in the presence of strong refraction or superrefraction. Because strong refraction reduces the uncertainties of refraction angle and impact parameter in the radio-optic analysis, these two methods can be combined.

The basic advantage of the back propagation method is its high accuracy in the reconstruction of the profile of refraction angle when it is possible to position the back propagation plane in a single-ray area. However, even when it is impossible, this situation can at least be diagnosed by ambiguities in the dependence $\varepsilon(p)$ or by severe amplitude fluctuations in the back propagation plane.

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